REGRESS(): The game of the General Linear Model

By Stephen Politzer-Ahles

Object of the game

You and your opponent(s) each get a cloud of data points. Your goal is to draw a better-fitting regression line through your data points than your opponent(s) do theirs. You will take turns adjusting your own line (or interfering with your opponent's line) little by little until one of you gets a nice-fitting line or you collectively run out of cards. The person with the best-fitting regression line wins.

Game setup

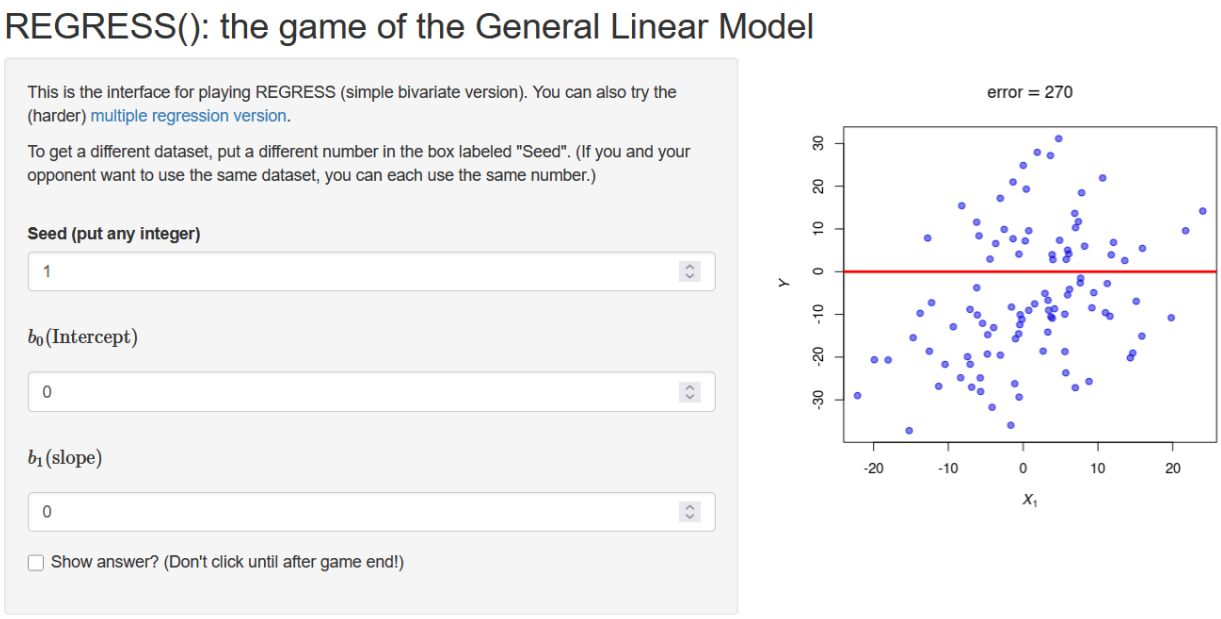
Set the face-down deck of cards in the middle of the table. Deal 3 cards to each player to be their starting hand. Deal the five top cards of the remaining deck and lay them out in a row face-up as a "card market". Place an index card on top of the remaining deck just so people can't see what's on top (because the paper I printed on is a bit see-through).

Each player should have their own computer (or some other kind of device), and sit in a way such that players can see each other's screens.

Each player open https://politzerahles.shinyapps.io/regress-simple/ on their computer.

(For a more advanced game, instead open https://politzerahles.shinyapps.io/regress-multiple/. In that app, choose "1 categorical and 1 continuous" for a medium-advanced game, or "2 continuous" for an advanced-advanced game.)

Each player will see an interface like this:



Each player should put some integer in to the field labeled "Seed"; this will give them a different scatter of data. (If players want to compete over the *same* dataset, they can enter the same seed.)

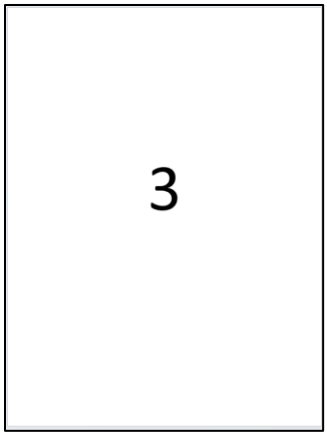
Do not click the "Show answer" button!

Shuffle the deck of cards and deal each player a hand of seven cards.

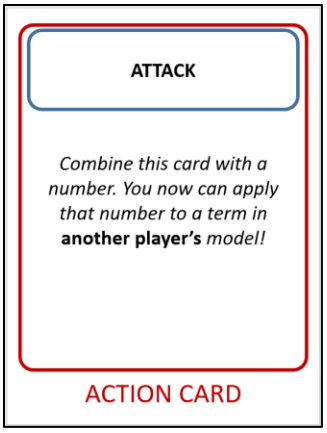
The player whose birthday is closest to a Fibonacci number (1, 2, 3, 5, 8, 13, 21, 34) goes first. In case of a tie, the tied player with the lower birthday goes first.

Cards

There are two kinds of cards: number cards, and action cards.



Sample Number card



Sample Action card

Number cards

Number cards just have a number on them. On your turn, you can play a number card to increase one of your model terms (in the simple regression model, either the intercept or the slope) by that amount. For example, the game starts with the slope (b_1) set to zero, so if you play a "3" you increase the slope from 0 to 3; when you put a "3" in the app it will automatically update the graph. On your next turn, playing a "1" would increase the slope from 3 to 4. Playing a "-1", on the other hand, would have changed the slope from 3 to 2 (an "increase" of -1 is actually a decrease!)

Action cards

Action cards combine with number to have some special effect. There are four kinds of action cards in the deck:

- MULTIPLY(): You can combine this card with up two number cards to treat them as their product. For example, if you play MULTIPLY() together with a 3 and a 2, you can treat that number as a "6" and add it to one of your model terms.
- NEGATIVE(): You can combine this card with a number card to multiply that number card by -1. For example, playing NEGATIVE() together with a 0.5 yields "-0.5".
- DUPLICATE(): You combine this card with a number card (or a "virtual" number card created by using MULTIPLY()) to use that number twice. You can either use it twice on the same term (e.g., adding 3 twice to the same term is the same as adding 6), or use it on two different terms (e.g., on the slope and on the intercept).
- ATTACK(): You combine this card with any number card other than a 5 to add that number to a term in *someone else's* model. So you can use this to mess up someone else's carefully-built model! (If you want a friendlier game, you can play without this card.)

You may use up to one of each action card together – for example, you can combine ATTACK() and MULTIPLY() with number cards to create ATTACK(MULTIPLY(2, 3)) to add a 6 to a term in an opponent's model.

Order of play

On your turn, you can do any **one** of the following things:

- Draw *one* face-up card from the card market, and replace it with a new card from the deck;
- Draw *two* face-down cards from the deck;
- Discard any number of cards from your hand to draw that many face-down cards from the card market; or
- Use a number card (optionally combined with one or more action cards) to add that value a term in a regression model. Put the new regression term into the app and watch the regression line change. (Note, you have to do the addition in your head. For example, if the Intercept is currently 3 and you want to add 1 to it, you can't just enter 1 in the app and have it update; you need to mentally calculate that 3+1=4, then you put 4 into the app.)

Whenever you update a regression model, you will see the "MSE" number at the top of the graph change. MSE is short for Mean Squared Error; it's a measure of how badly your line fits through the dots. Lower MSE means a better-fitting line. You're trying to get your MSE as low as possible.

Game end

The game ends immediately when **either** of the following things happens:

- A player's gets their MSE lower than the goal MSE; or
- The last card is drawn

The player with the lowest MSE is the winner.

After the game is over, you can click the "Show answer" button, if you want, to see the coefficients for the best-fitting regression model.

Variant: Coopetition mode

Another way to play the game (and possibly more conducive to a whole-class discussion) is to have the whole group work together to fit a regression line to the same dataset, while simultaneously competing to see who can best predict how well the regression line fits.

To play this way, remove the cards from the game, and select the "Coopetition mode" option shown in the game interface. This will show a row of random cards at the bottom of the interface, which will refresh every time you update the model. This is now the market of cards that players can use.

In the game, players take turns to use cards to update the model, as in the normal game. But after a player chooses a card, ask the whole group to write down their best guess of what the MSE will change to. After everyone has written down their guess, update the model and see how the MSE changes. Everyone can then calculate how far their guess was from the actual MSE: abs([guessed MSE] – [actual MSE]). At the end of the game, the player(s) with the lowest average difference-from-actual across rounds is(are) the winner(s).

(This mode is particularly fun if all players track these on a shared spreadsheet, which can also include a separate tab which compares everyone's difference-from-actual round by round. Since players can see each other's guesses and see who is winning as the game goes on, this encourages friendly competition, while also getting players to talk about what's happening in the game. The game website includes an example of the spreadsheet I used for one iteration of this game in class.)

Teaching notes

The purpose of this game is not actually to teach people how to fit a regression line through data; statistical software does that for them. The purpose is, rather, to help them think about what regression coefficients mean. The core mechanic of the game (choosing cards to change the line) follows from that.

Players sometimes might already be able to tell what the coefficients *should* be (e.g., they might already be able to see that the slope should be about 5 and the intercept should be about 10), but they have to deal with the cards available to them, which might not be cards that get to those values. In extreme circumstances, they might not even have any cards that can improve the model (e.g., they might know that the slope needs to get smaller, but they

have no negative cards, and therefore any card they play will make the model *worse*); in this circumstance all they can do is figure out what will do the least harm.

This is all by design. Interpreting regression models isn't actually about fitting the line; it's about understanding what differences in coefficients mean (e.g., looking at an interaction coefficient of "2" and realizing "that makes this group's slope be 2 point steeper than the other group's slope", if that's what makes sense given how the model was coded). The game is meant to let people practice this sort of thinking.

Therefore, I encourage playing with the cards, even if they sometimes feel frustrating when players see where they want the line to go and just don't have the cards to send it there. After the game is done, you can have a reflection/debrief section talking about the skills you hope players got to practice during the game, and highlight this point.

Copying the game

I cannot guarantee that my Shiny app interface for this game will always be available. If you want to play it, I fully support you making your own copy. If you make your own copy you can also modify things if you want the game to work differently.

A .zip file of the game code is provided at the game website. You can download and unzip that file and then follow the instructions at Getting started with shinyapps.io to host your own copy of it.

Acknowledgements

This game was inspired by a game about library searching that Kelly Hangauer and Carmen Orth-Alfie designed and let me play in summer 2026. I thank Samantha Demmerle and Joshua Potter for playtesting and feedback, and students in my fall 2026 "Regression and mixed-effects modeling for language science" course for being my guinea pigs. Finally, I like to think this game owes an intellectual and spiritual debt to the granddaddy of all gamified math learning experiences: the inimitable *Math Blaster!*.